

Zambia University College of Technology

Advancing Knowledge and Innovation

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COURSE: Operating System Concepts

PROGRAM: Bachelor of Cybersecurity

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Operating System Installation

1. Introduction

An operating system is a system software that manages computer hardware and provides an environment in which application software can operate. Examples of popular operating systems include, MacOS, Linux based OS and Microsoft Windows.
For this practical assignment, I chose Kali linux as the operating system to be install. Kali linux is a Debian based linux distribution developed for Cybersecurity, penetration testing, digital forensics, and security auditing.
The installation was performed on a virtual machine using QEMU/KVM. Virtualization allows an operating system to run inside an isolated virtual computer while sharing the physical resouces fo the host machine.
The purpose for this practical assignment was to demonstrate the process of deploying a Linux based operating system in a virtualized environment, configuring it’s resources and network connectivity and making sure the installation was successful.

2. Objectives

- Install a Linux based operating system on a virtual machine.
- Allocate virtual hardware resources.
- Configure networking
- Boot and Test basic system functionality
- Verify the operating system’s hardware and network connectivity
- Document the installation process and configuration

3. Operating System and System requirements

The following operating system and tools were used during the installation

Component	Details
Guest operating system	Kali Linux
Host operating system	Nobara Os (Fedora based)
Virtualization platform	QEMU/KVM
Virtual machine manager	Virt-manager
Installation media	Kali Linux ISO
Architecture	x86_64
Virtual CPU	2 cores
Virtual RAM	4GB
Virtual storage	25GB
Network	Virtual NAT network

4. Creating the virtual machine

The Kali linux installation ISO was downloaded from the official Kali Linux website. Before installing Kali , the virtual machine was configured with the recommended resources. The ISO image was used as the bootable installation media for the virtual machine, unlike a prebuilt virtual machine image, an ISO installation allows the operating system to be installed from the beginning, including disk partitioning, user creation, package installation, and bootloader configuration. And Kali's official QEMU/libvirt documentation demonstrate the use of virt-manager with local ISO and recommends a configuration of at least 2GB RAM and 2 cores. During the virtual machine creation, kali linux was not listed as a specific distribution in the available operating system profiles, since Kali linux is a Debian based distribution the closest option I chose was Debian 13. The selection of the specific linux distribution is only to provide the virtualization software with the appropriate guest operating system defaults and does not change the kali linux installation media.

4.1 QEMU/KVM Virtual machine configuration

figure 1: Linux distribution and Virtual memory allocation and CPU

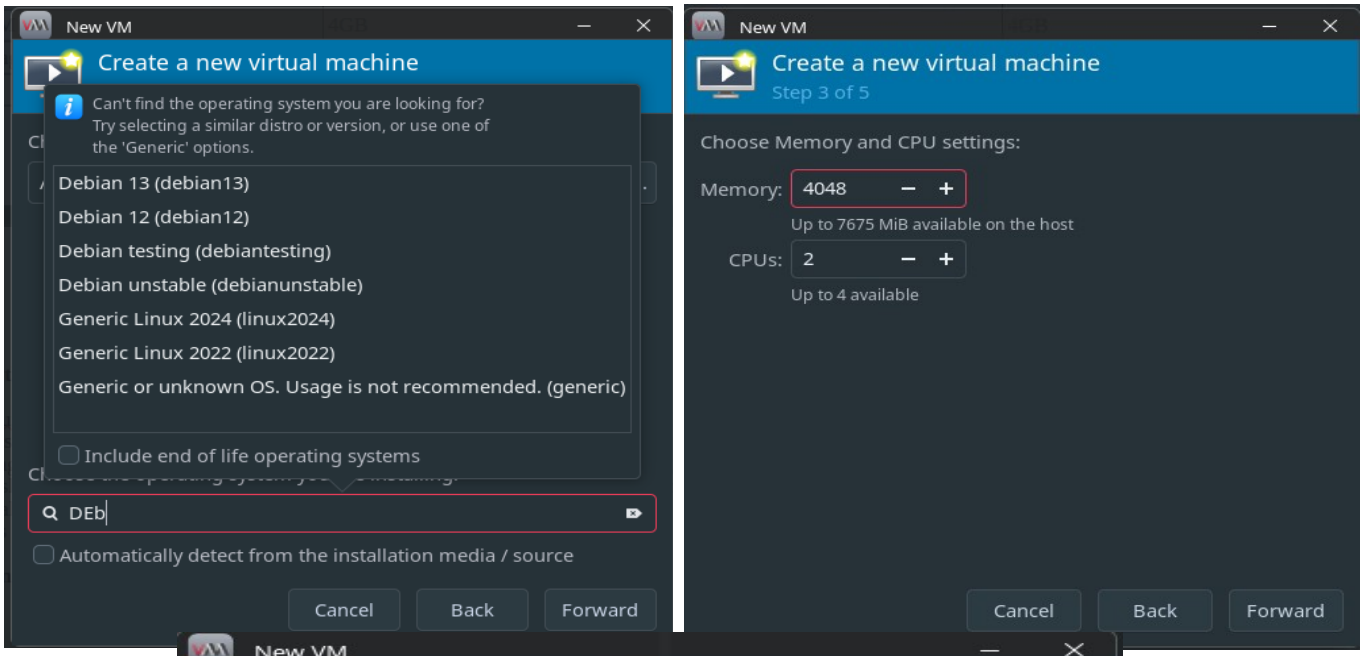


figure 2: Virtual disk configuration

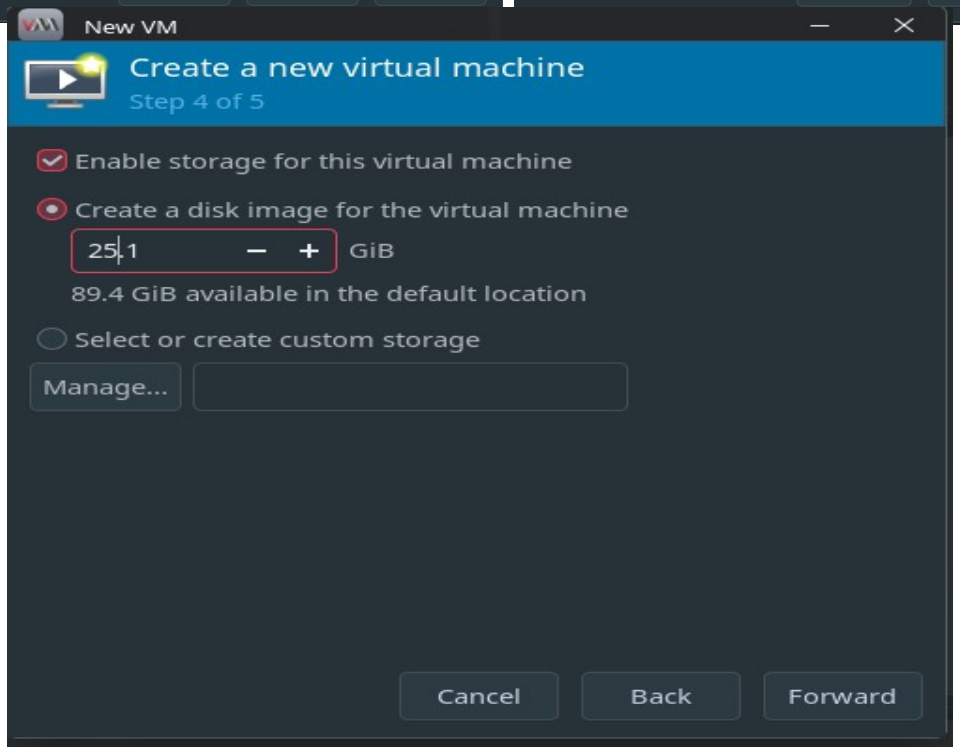
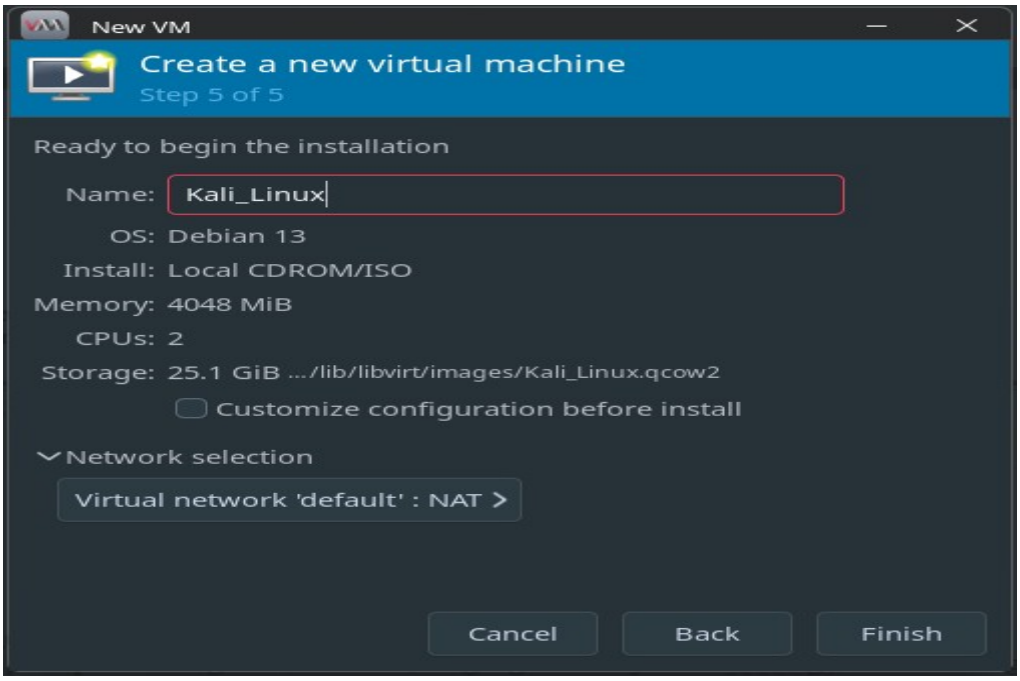


figure 3: Virtual network configuration and resources allocated



6. Installation Process

6.1 Booting the Kali Linux installer

After the virtual machine was created, it started with the kali limux ISO attached as the boot device. Kali linux installation menu appeared, providing different installation options.

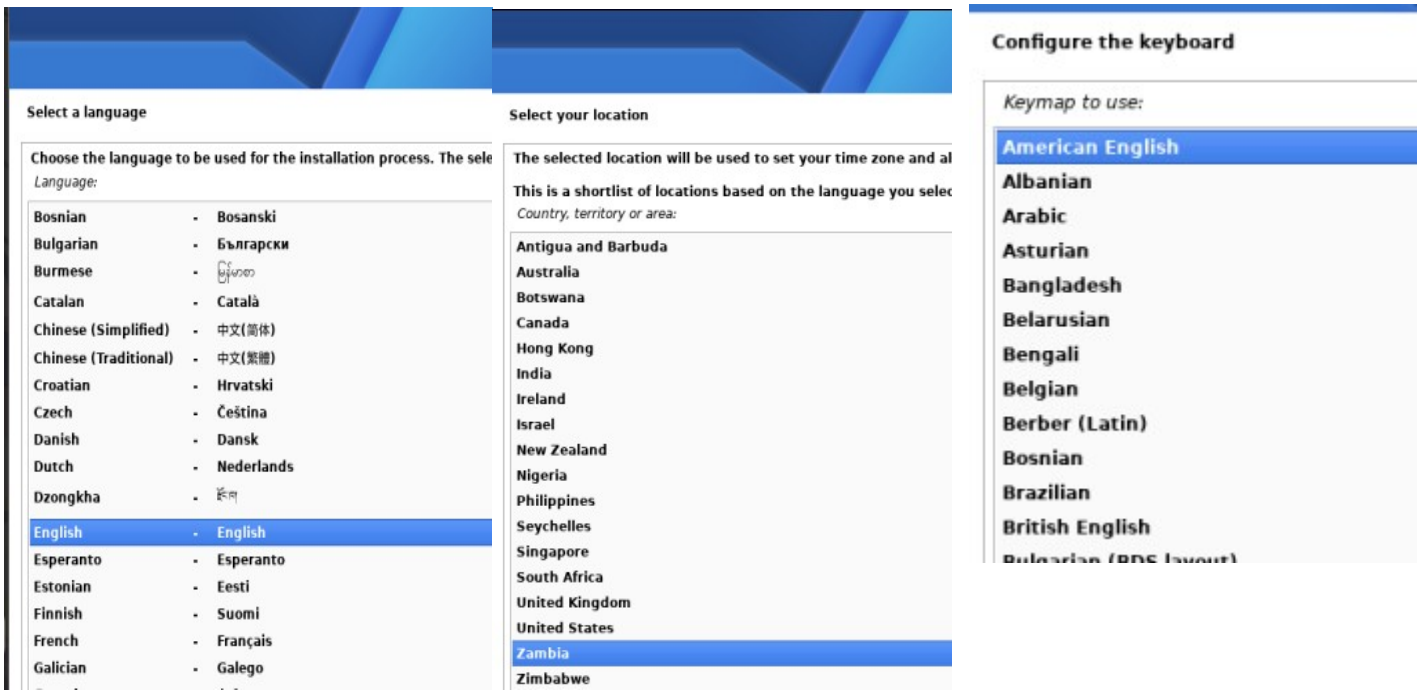
Figure 1: The graphical installation was selected.



6.2 Language, Location and Keyboard configuraation

The installation media required that the language, geographical location and Keyboard to be set. The appropriate language and location according to the stand, the keyboard layout was then configured according to the keyboard being used on the host machine.

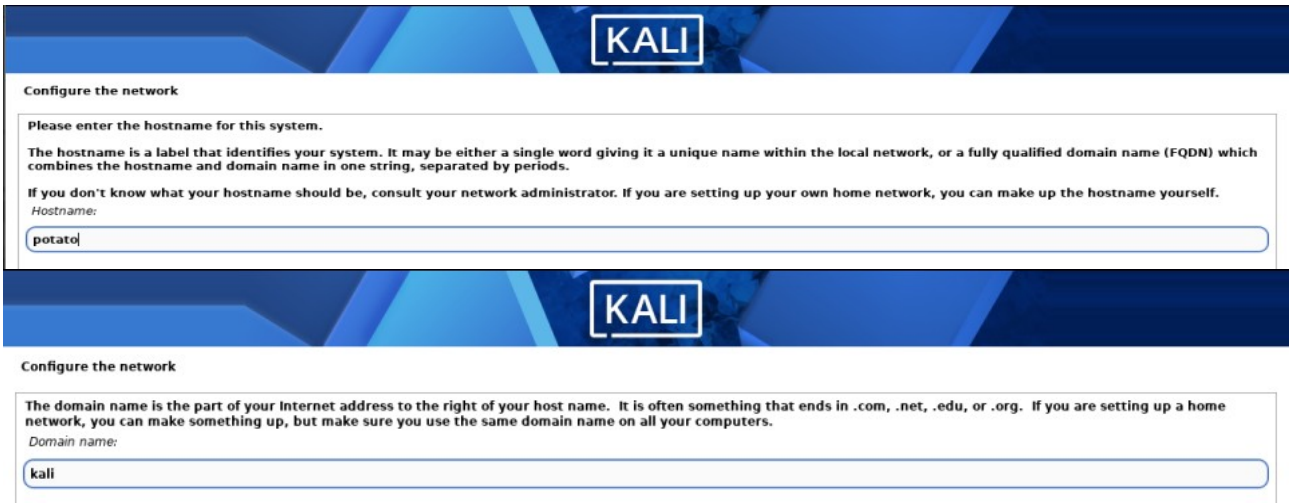
figure 1: Language, Location and keyboard



6.3 Network Configuration

The installer detected the virtual network interface and obtained the network configuration automatically. Kali linux's documentation describes the installer probing the network interface and using DHCP where available and since my home network was already configured with DHCP the operating system automatically received an IP address. The hostname and domain name were configured for the operating system.

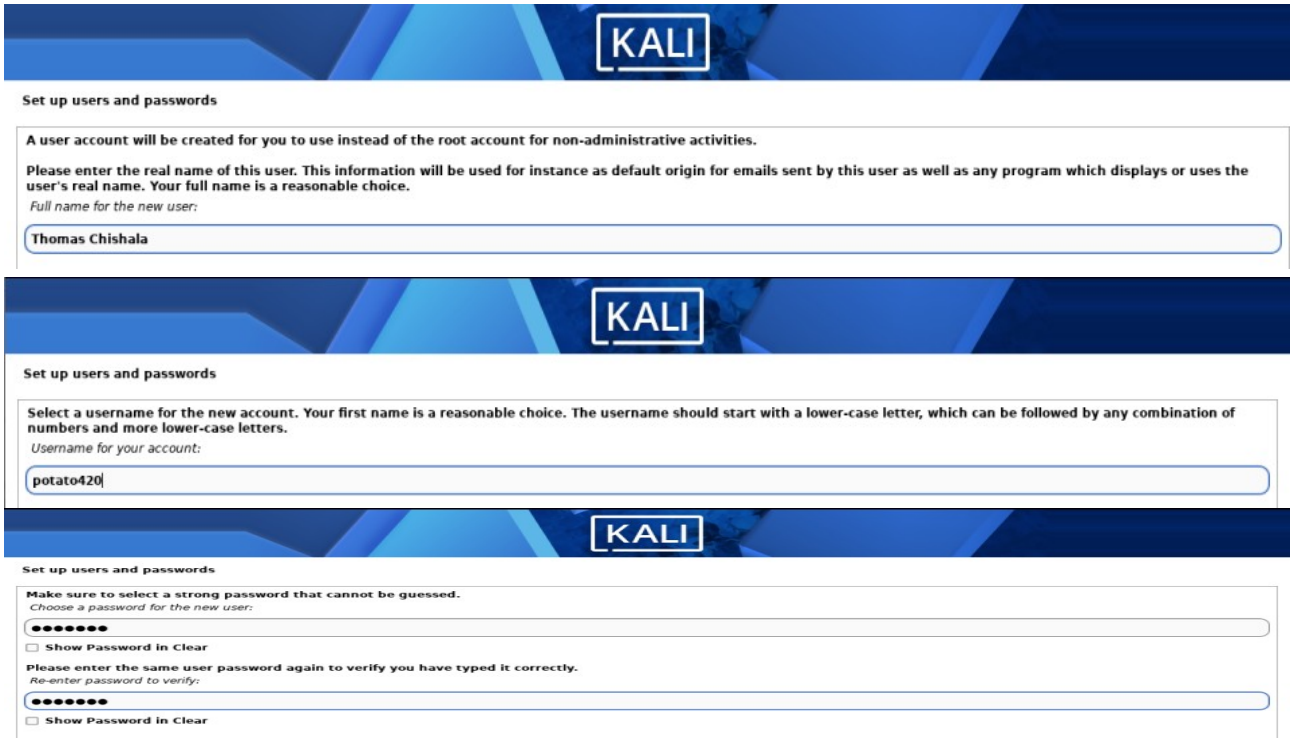
Figure 1: Network configuration



6.4. User account configuration

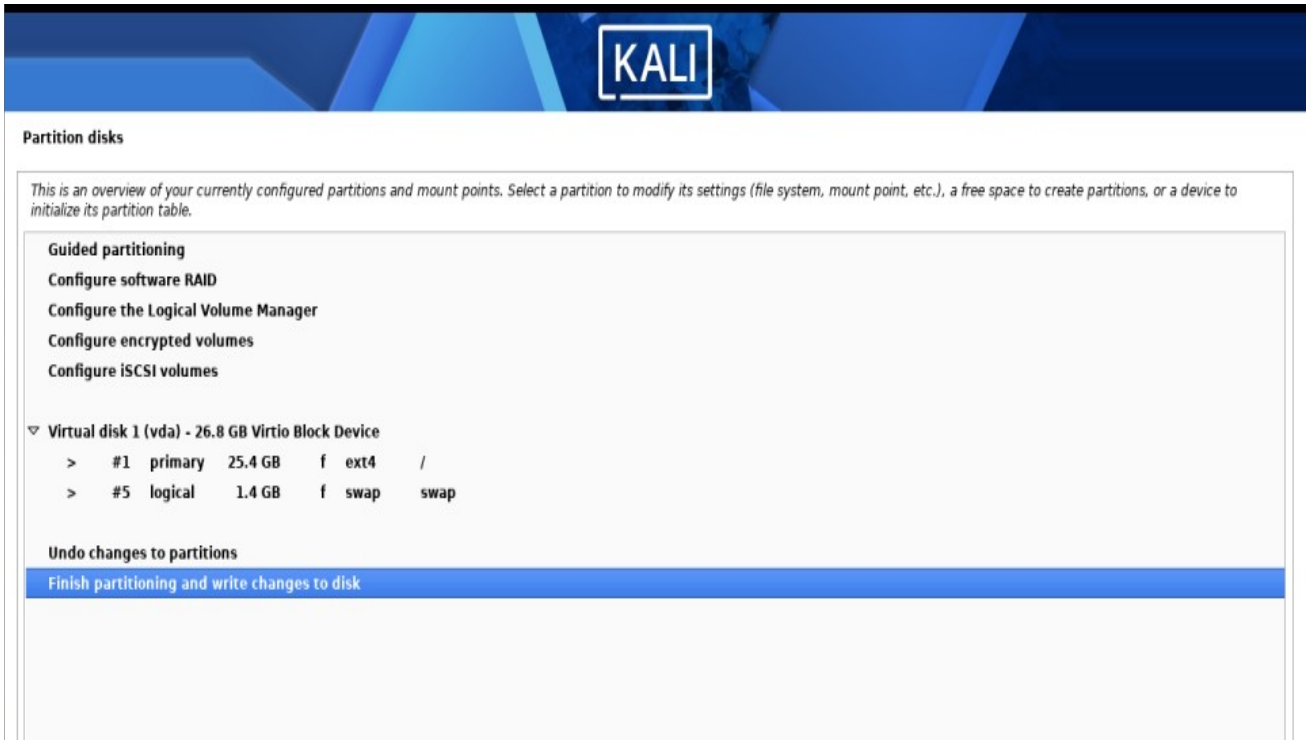
A user account was created during the installation process, .which included my full name, username and a password.

figure 1: user account configuration



6.5 Disk Partitioning
The guided partitioning option was selected because the virtual machine contained a dedicated virtual disk intended for the operating system installation.
The virtual disk was partitioned and formatted before the operating system files were coped on to it

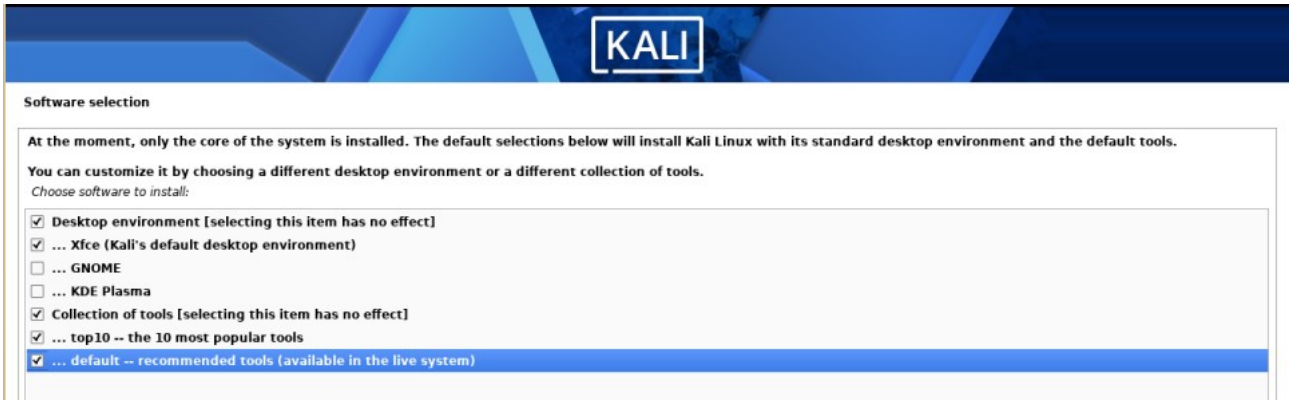
figure 1: Disk partitioning configuration



6.6 Software Installation

After the disk was configured, the installer copied all the required operating system files to the virtual disk. The required packages and desktop environment were installed. I decided to go with the default selection of the packages and desktop environment.

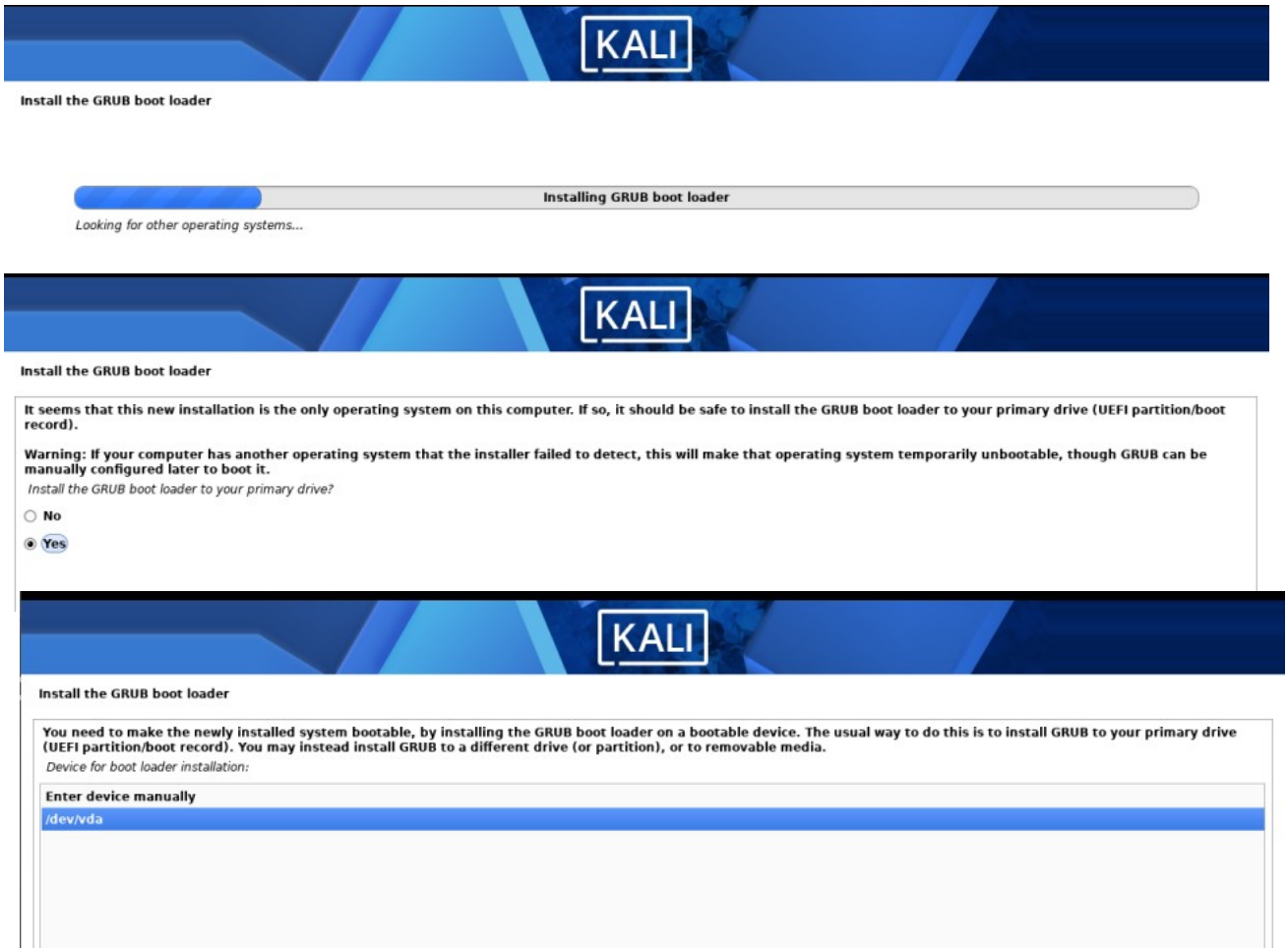
Figure 1: software installation process



6.7 Bootloader Installation

The GRUB bootloader was installed on the virtual disk. The bootloader is important because it is responsible for loading the installed operating system when the virtual machine starts. After the grub was installed, the virtual machine was restarted.

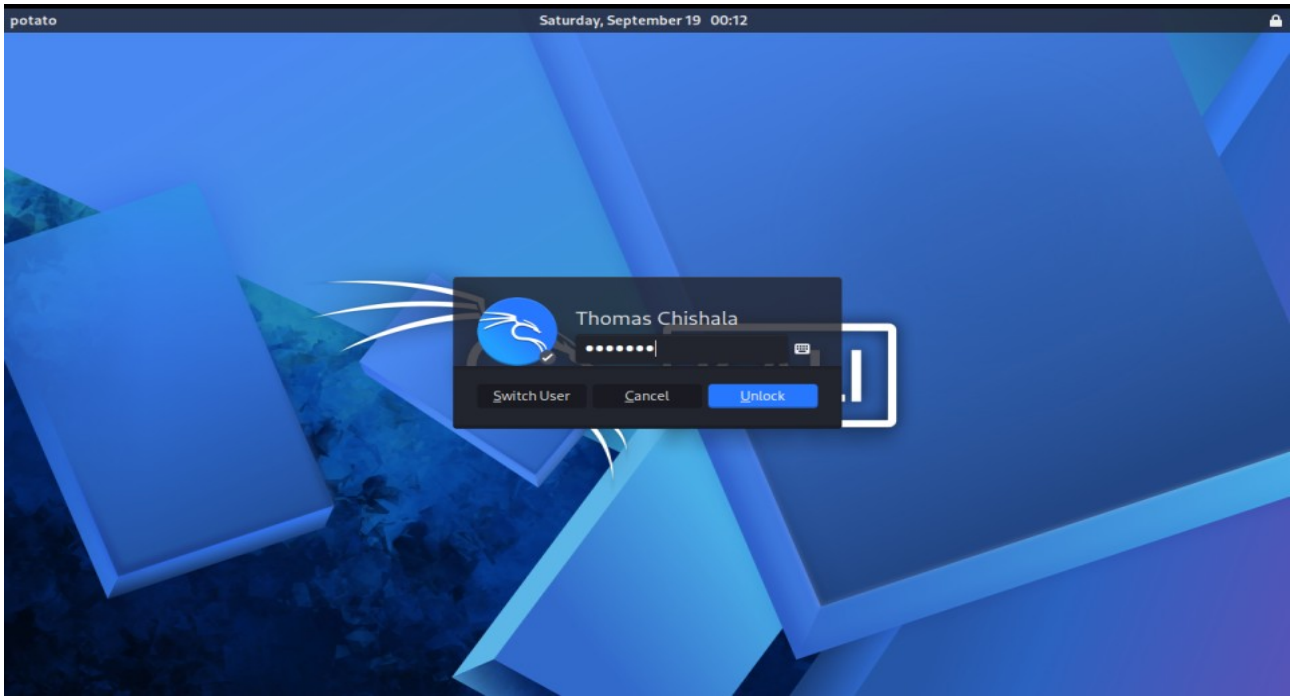
Figure 1: GRUB Installation



7. Initial Configuration after installation

After restating the virtual machine, Kali linux successfully booted form the virtual disk. I then used the user account created during the installation .

Figure 1: Kali linux booted.

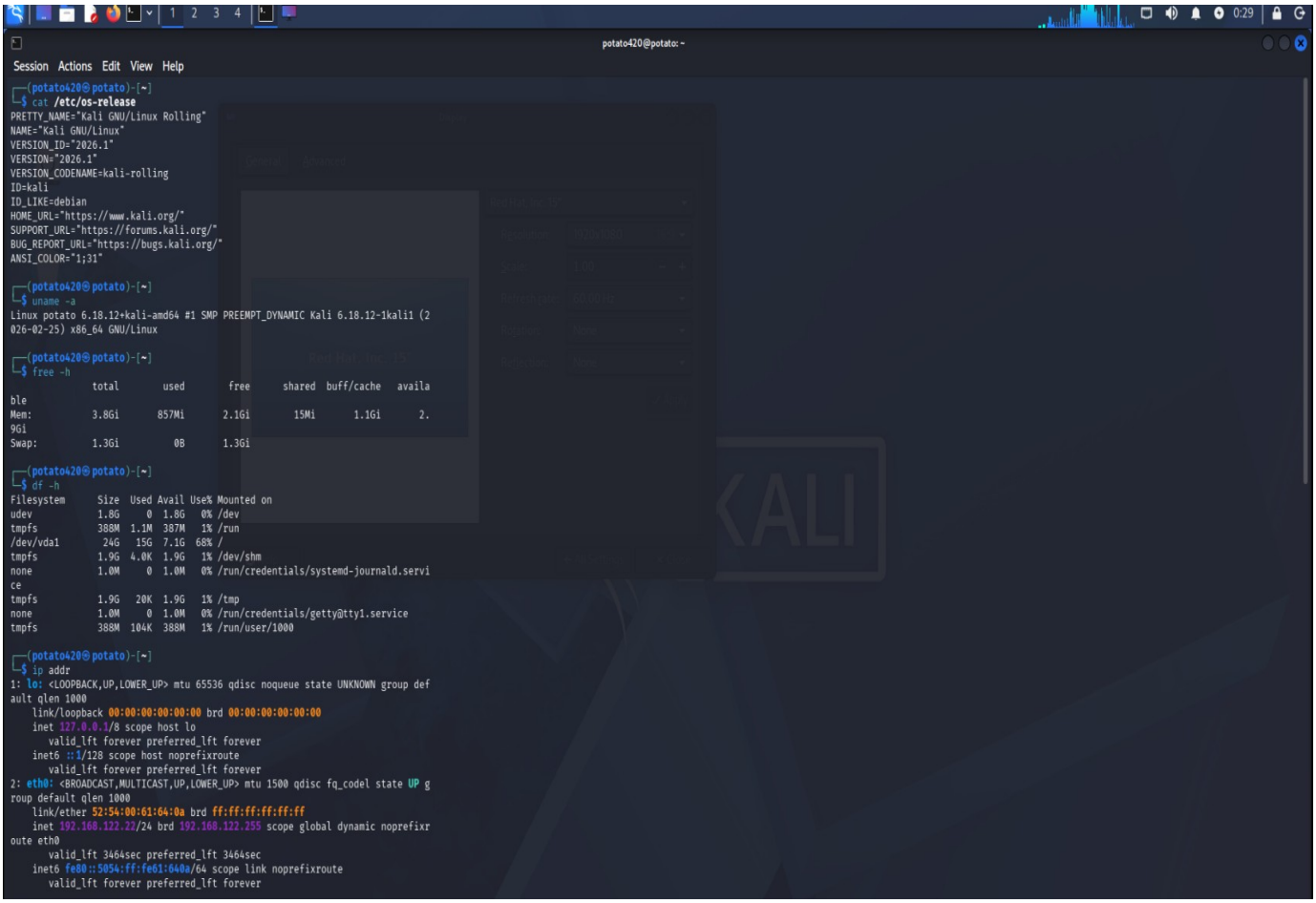


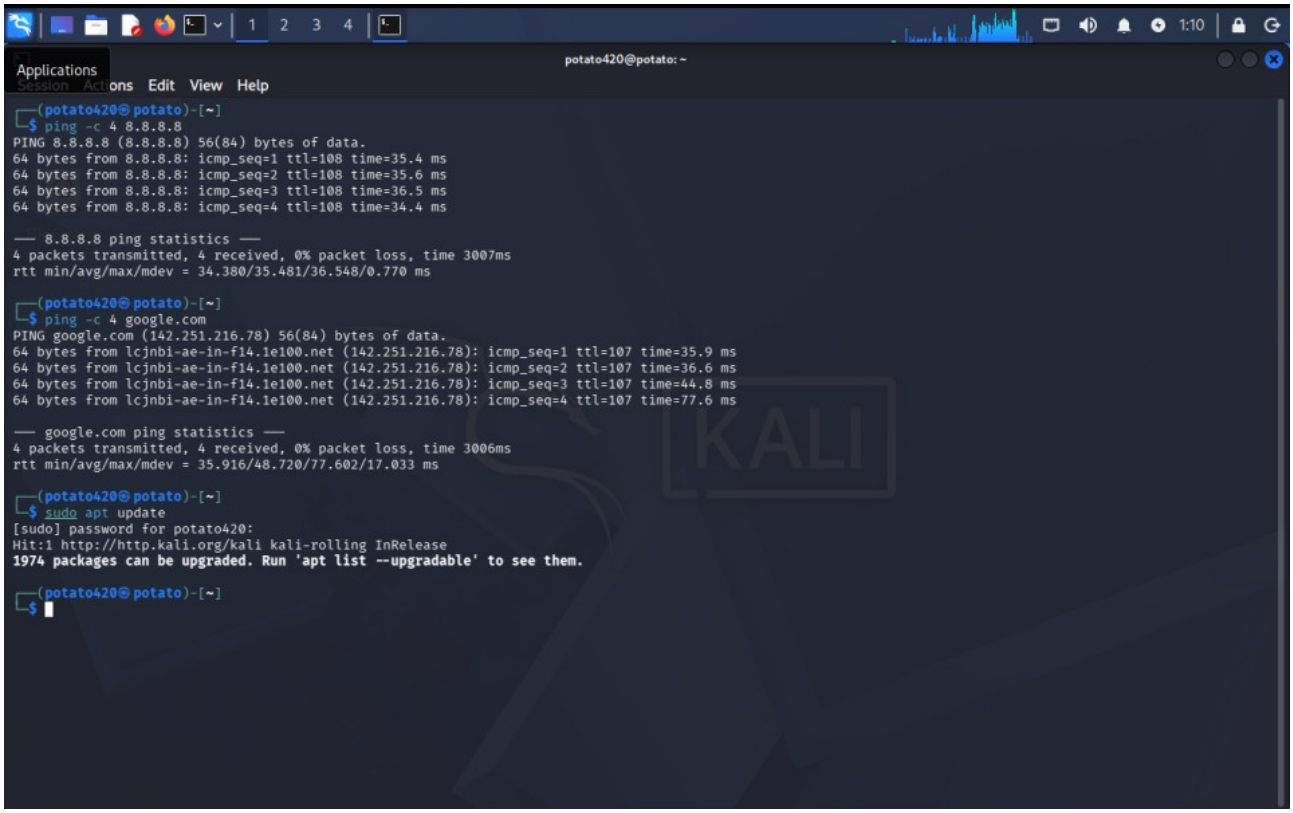


7.1 Checking the operating system

The following the commands were used:

- cat /etc/os-release: Displays information about the operating system
- uname -a: Displays information about the linux kernal
- free -h: Checks the amount of memory recognized by the operating system
- df -h: Displays filesystem usage and available storage
- ip addr: Shows the ip address of the linux network interface and assigned address.
- Ping -c 4 8.8.8.8: The command sends four ICMP packages to destination
- ping -c 4 google.com: Domain name resolution
- sudo apt update: package repository was tested.





8. Conclusion

The operating system was installed and verified successfully. The process involved identifying the system requirements, creating and configuring the virtual machine, attaching the Kali limux ISO, performing the operating system installation, configuring the installed system, and testing it's functionality. The post installation tests confirmed that Kalil linux successfully recognized the virtual hardware. The assignment demonstrated the usfulness of virtualization for an opeerating system system can be installed, configured, tested and isoated without replacing the host operating system.

9. References

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